

$$ATT_{or,rc}^{ny}(g, t; \delta) = \mathbb{E}_M \left[\frac{G_g}{\mathbb{E}_M[G_g]} \left(\left(m_{g,t}^{rc,treat}(X) - m_{g,g-\delta-1}^{rc,treat}(X) \right) - \left(m_{t+\delta,t}^{rc,ny}(X) - m_{t+\delta,g-\delta-1}^{rc,ny}(X) \right) \right) \right],$$

the inverse probability weighted (IPW) estimands

$$\begin{aligned} ATT_{ipw,rc}^{nev}(g, t; \delta) &= \mathbb{E}_M \left[(w^{treat}(g, t) - w^{treat}(g, g - \delta - 1)) \cdot Y \right] \\ &\quad - \mathbb{E}_M \left[(w_{nev}^{comp}(g, t) - w_{nev}^{comp}(g, g - \delta - 1)) \cdot Y \right], \\ ATT_{ipw,rc}^{ny}(g, t; \delta) &= \mathbb{E}_M \left[(w^{treat}(g, t) - w^{treat}(g, g - \delta - 1)) \cdot Y \right] \\ &\quad - \mathbb{E}_M \left[(w_{ny}^{comp}(g, t, t + \delta) - w_{ny}^{comp}(g, g - \delta - 1, t + \delta)) \cdot Y \right], \end{aligned}$$

and the doubly-robust (DR) estimands

$$\begin{aligned} ATT_{dr,rc}^{nev}(g, t; \delta) &= \mathbb{E}_M \left[\frac{G_g}{\mathbb{E}_M[G_g]} \left(\left(m_{g,t}^{rc,treat}(X) - m_{g,g-\delta-1}^{rc,treat}(X) \right) - \left(m_{c,t}^{rc,nev}(X) - m_{c,g-\delta-1}^{rc,nev}(X) \right) \right) \right] \\ &\quad + \mathbb{E}_M \left[w^{treat}(g, t) (Y - m_{g,t}^{rc,treat}(X)) - w^{treat}(g, g - \delta - 1) (Y - m_{g,g-\delta-1}^{rc,treat}(X)) \right] \\ &\quad - \mathbb{E}_M \left[w_{nev}^{comp}(g, t) (Y - m_{c,t}^{rc,nev}(X)) - w_{nev}^{comp}(g, g - \delta - 1) (Y - m_{c,g-\delta-1}^{rc,nev}(X)) \right], \\ ATT_{dr,rc}^{ny}(g, t; \delta) &= \mathbb{E}_M \left[\frac{G_g}{\mathbb{E}_M[G_g]} \left(\left(m_{g,t}^{rc,treat}(X) - m_{g,g-\delta-1}^{rc,treat}(X) \right) - \left(m_{t+\delta,t}^{rc,ny}(X) - m_{t+\delta,g-\delta-1}^{rc,ny}(X) \right) \right) \right] \\ &\quad + \mathbb{E}_M \left[w^{treat}(g, t) (Y - m_{g,t}^{rc,treat}(X)) - w^{treat}(g, g - \delta - 1) (Y - m_{g,g-\delta-1}^{rc,treat}(X)) \right] \\ &\quad - \mathbb{E}_M \left[w_{ny}^{comp}(g, t, t + \delta) (Y - m_{t+\delta,t}^{rc,ny}(X)) - w_{ny}^{comp}(g, g - \delta - 1, t + \delta) (Y - m_{t+\delta,g-\delta-1}^{rc,ny}(X)) \right]. \end{aligned}$$

The OR, IPW and DR estimands respectively generalize [Heckman et al. \(1997\)](#), [Abadie \(2005\)](#) and [Sant'Anna and Zhao \(2020\)](#) estimands for the two groups, two periods DiD setup to the staggered DiD setup with multiple periods and multiple groups.

Theorem B.1. *Let Assumptions 1, 3, 6, and B.1 hold.*

(i) *If Assumption 4 in the main text holds, then, for all g and t such that $g \in \mathcal{G}_\delta$, $t \in \{2, \dots, \mathcal{T} - \delta\}$ and $t \geq g - \delta$,*

$$ATT(g, t) = ATT_{ipw,rc}^{nev}(g, t; \delta) = ATT_{or,rc}^{nev}(g, t; \delta) = ATT_{dr,rc}^{nev}(g, t; \delta).$$

(ii) *If Assumption 5 in the main text holds, then, for all g and t such that $g \in \mathcal{G}_\delta$, $t \in \{2, \dots, \mathcal{T} - \delta\}$ and $t \geq g - \delta$,*

$$ATT(g, t) = ATT_{ipw,rc}^{ny}(g, t; \delta) = ATT_{or,rc}^{ny}(g, t; \delta) = ATT_{dr,rc}^{ny}(g, t; \delta).$$

We defer the proof of Theorem B.1 to the Supplementary Appendix. The identification results in Theorem B.1 suggest a simple two-step estimation procedure for the $ATT(g, t)$ with repeated cross-section data that is analogous to the panel data case discussed in Section 4. The asymptotic properties of such two-step estimators follow from analogous arguments; the details are omitted for brevity. Likewise, we can aggregate these estimators to provide summary measures of the causal effects like those discussed in Section 3.

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